Data Flow Diagram

Date	30 September 2026
Team ID	SWTID-2026-5240
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	2 Marks

Data Flow Diagram Components & Description

1. External Entities (Sources / Users):

End User / Requester:

Role: Submits a service request for a "Standard Laptop" via the ServiceNow Service Catalog portal.

IT Manager / Approver:*

Role: Reviews and approves the requested item (RITM) record in ServiceNow.

Hardware Assignment Group / Technician:

Role: Receives automatically generated catalog tasks (sc_task) to configure and deliver the hardware.

2. Processes (Workflow Operations):

Process 1.0 (Submit & Create Service Request):

Description: Receives request details from the user, generates the Request (REQ) and Requested Item (RITM) records, and triggers the approval process.

Process 2.0 (Trigger Flow & Evaluate Approval):

Description: Listens for approval events on the Requested Item (RITM) and triggers the Standard Laptop Task flow in Flow Designer.

Process 3.0 (Generate & Assign Catalog Task):

Description: Executes the Create Catalog Task action, sets the short description to "Laptop need to Configured", and assigns the task to the Hardware group.

3. Data Stores (Tables & Engines):

Data Store D1 (Service Catalog Item - sc_cat_item):

Description: Holds catalog item specifications, forms, and process engine configurations for the Standard Laptop.

Data Store D2 (Requested Item Table - sc_req_item / RITM):

Description: Stores individual ordered item records, approval statuses, and requester details.

Data Store D3 (Catalog Task Table - sc_task):

Description: Stores actionable fulfillment tasks assigned to the Hardware assignment group.

4. Data Flows (Movement of Data):

Flow 1: End User → Process 1.0 (Submits Standard Laptop Order)

Flow 2: Process 1.0 → Data Store D1 (Fetches Catalog Item Details)

Flow 3: Process 1.0 → Data Store D2 (Creates RITM Record in Requested Item Table)

Flow 4: IT Manager / Approver → Process 2.0 (Approves Requested Item)

Flow 5: Data Store D2 → Process 2.0 (Triggers 'Standard Laptop Task' Flow)

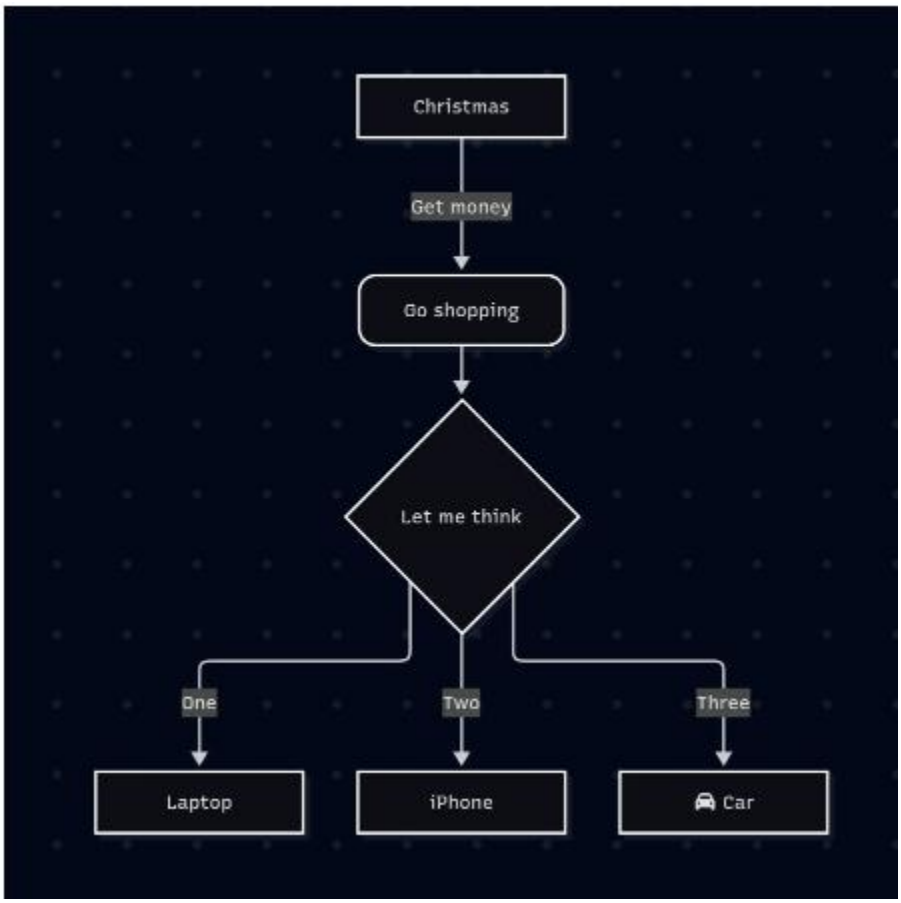
Flow 6: Process 2.0 → Process 3.0 (Executes Flow Action: Create Catalog Task)

Flow 7: Process 3.0 → Data Store D3 (Saves Generated Task assigned to Hardware Group)

Flow 8: Data Store D3 → Hardware Assignment Group (Routes Task for Laptop Configuration)DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example


Christmas
Get money
Go shopping
Let me think
One
Two
Three
Laptop
iPhone
Car